

Li-ion battery state of health prediction through metaheuristic algorithms and genetic programming

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ABSTRACT

Predicting the State of Health (SOH) of the Lithium-ion battery with higher accuracy and reduced cost is a challenging and crucial task for ensuring its reliability and safety. To achieve this, a two-stage prediction framework is proposed to find a concise expression of SOH using the health features of the battery through metaheuristic algorithms and genetic programming (GP). In Stage-I, three conflicting objectives are considered concurrently: the root-mean-square error (RMSE) of SOH prediction, the health features selected, and the expressional complexity of the battery capacity. The wrapper structure is utilized for SOH prediction, where a binary multi-objective grey wolf optimization (Binary MOGWO) algorithm is employed to select features and generate the Pareto set. Genetic programming is used to calculate the SOH value using symbolic regression models. In Stage-II, the final compromise solution is filtered from the Pareto set through decision-making approaches. The relationships between selected features and the capacity of the battery are investigated. The NASA Prognostics Center of Excellence battery dataset is chosen to verify the effectiveness of the proposed framework. The simulation results show that the features found through the framework can provide SOH predictions in the form of symbolic equations with higher accuracy, lower cost, and reduced complexity.

1. Introduction

Li-ion batteries are extensively used in various fields such as consumer electronics, electric vehicles, aerospace satellites, unmanned aerial vehicles and shipbuilding due to their advantages in energy density, lifetime, and power capacity (Yang et al., 2021; Chunwu et al., 2023). Battery performance gradually declines with cyclic charging and discharging operations. The trajectory of this trend is measured by the State of Health (SOH) value, which is an artificial measurement indicating a point in the battery's lifetime and evaluating the health level of its current performance compared to its fresh state (Yang et al., 2021). Accurate SOH prediction is an important task and has been studied by many researchers (Berecibar et al., 2016; Fan et al., 2020).

SOH prediction methods can be roughly classified into traditional methods and emerging methods, where traditional methods encompass empirical and model-based approaches, and emerging methods predominantly involve advanced data-driven techniques (Wang et al., 2020; Vanem et al., 2021). The shift from traditional to emerging methods marks a significant advancement in SOH prediction capabilities. Traditional methods such as equivalent circuit models (ECMs) (Hu

et al., 2020) and electrochemical models (EMs) (Shu et al., 2021) provide foundational insights in theory but often lack flexibility due to their reliance on specific testing conditions. Both ECMs and EMs rely on assumptions and simplifications to make the models tractable. ECMs simplify the battery into a combination of resistors, capacitors, and voltage sources, which may not capture all the nuances of battery behavior under different conditions. Similarly, EMs, while more detailed, still require assumptions about the battery's internal processes, which may not hold true under all operating scenarios (Hu et al., 2020; Shu et al., 2021).

Emerging methods leverage data-driven techniques to capture complex, non-linear relationships in the data, leading to higher prediction accuracy and adaptability to varying operational conditions (Oji et al., 2021). Frequently used data-driven methods for SOH include artificial neural networks (ANNs), Support Vector Machines (SVMs), Gaussian process regression (GPR), relevance vector machines (RVMs), Elastic Nets, Random Forests, and others (Fei et al., 2021; Hu et al., 2021a). These models, especially machine learning and deep learning approaches, can identify and learn intricate patterns and dependencies in the data that traditional empirical models might miss, resulting in

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more accurate predictions.

However, these advancements also bring challenges. The complexity of machine learning models necessitates substantial computational resources and high-quality data for training. Additionally, the interpretability of deep learning models remains an area of ongoing research, as their "black-box" nature can obscure the underlying decision-making processes.

Genetic programming (GP) is a type of machine learning technique that is an evolutionary algorithm-based methodology inspired by biological evolution and natural selection mechanisms, much like genetic algorithms (GA) (Gandomi et al., 2015a). However, its population consists of computer programs stored as tree structures. The objective of GP is to optimize these programs to accomplish specific computation tasks, such as classification and regression. Unlike other ML methods, GP equations can use mathematical models to minimize the difference between real (observed) and estimated data. It is widely used in various engineering applications, such as image classification, ground motion prediction, and chemical engineering (Gandomi et al., 2015a; Riolo et al., 2016).

In the lithium-ion battery SOH field, temperature and discharge rates of batteries in electric vehicles are selected to model capacity using GP in the work of Garg, Peng, Le, and others (Garg et al., 2018). Mechanical parameters such as stack stress are also added to model the capacity of the battery based on GP (Shui et al., 2018). Several GP models are constructed for estimating the SOH of lithium-ion batteries using different features, aiming to minimize the root square error (Yao et al., 2020). Three objectives, residuals of voltage and temperature, and differences between the derivatives of the predicted and measured voltages are selected to be studied using grammar-based multiobjective genetic programming (MOGAP) (Echevarría et al., 2019), based on the feature points in the Open Circuit Potential (OCP) and Incremental Capacity Analysis (ICA) curves of the battery.

The explicit expressions of the GP results provide understandable expressions about SOH prediction. This characteristic of GP is different from other widely used machine learning methods, such as ANN and GPR (Zhao et al., 2023; Hu et al., 2021b), where all features are employed to enhance regression accuracy. However, the specific contribution of each feature is not clear due to their black-box performances.

Despite the recent developments of GP application in SOH study, there are still several research gaps to be filled. First, only specific features reflecting the aging performance of the battery are considered, rather than selecting and evaluating related features in the feature set. For example, five performance features (Yao et al., 2020) or peaks in ICA curves (Echevarría et al., 2019) are employed directly to construct capacity expressions. Many battery performance features have been developed (Hu et al., 2021a), and better expressions using other features with higher accuracy might be ignored just by selecting those features directly without further consideration.

Second, when GP is utilized for a SOH study, its main function is generating the capacity regression equation. The complexity of the expressions and the number of features selected are not considered as objectives at the same time. To achieve the necessary accuracy, much lengthier expressions are obtained if fewer features are selected (Garg et al., 2018; Shui et al., 2018). The understandability of the expression will be reduced then. It is anticipated that a better trade-off between expressional complexity and the number of selected features should be considered.

Lastly, after the expression of the capacity is acquired through GP, further analysis regarding why those performance features are selected is limited. More relationship analysis among features is also expected.

In this work, a wrapper method for SOH prediction is adopted, i.e., selecting features and finding the optimal feature subset based on a predefined ML prediction model (Fei et al., 2021). To obtain higher accuracy of SOH prediction and fewer features selected concurrently, a two-stage framework, composed of multiobjective optimization and

decision making, is proposed. In the first stage, after related performance features of the battery are collected, a binary multiobjective grey wolf optimization (Mirjalili et al., 2016) (Binary MOGWO) algorithm is employed to select features and construct the searching population. This population is sent to GP for generating capacity expressions. Three objective functions, i.e., the regression accuracy of this expression (e.g., RMSE), the expressional complexity, and the features selected of GP are sent back to Binary MOGWO. When the convergence condition is met, the Pareto set will be provided by Binary MOGWO.

In the next stage, one multi-attribute decision-making technique, 'nearest to the utopian solution' (Parsons, 2009), is utilized to filter the final compromise solution from the Pareto set according to the normalized Euclidean distance. The three objective functions of the utopian solution are composed of corresponding single-objective optimization results, which are obtained through the binary grey wolf optimization (Negi et al., 2020) (Binary GWO) algorithm. Preference-based decision-making methods (Ishizaka and Nemery, 2013) are also examined.

To gain further insight into the expression of SOH obtained from the above optimization and decision-making process, a multivariate analysis technique, Student's t-test (Gaggero and Martinelli, 2022; Esteco, 2017), is used to investigate the effect size of these features upon the capacity of the battery.

The main contributions and novelty aspects of this work are as follows:

- (1) A general framework, composed of multiobjective optimization and decision-making, is proposed for SOH prediction. The Pareto set is found through multiobjective optimization algorithms, followed by decision-making techniques to select the final compromise solution.
- (2) An illustrative application of the proposed framework is given. Binary MOGWO, GP, and decision-making techniques work together to find a concise expression of SOH with higher accuracy and lower cost.
- (3) A deeper understanding of the SOH expressional expression is mined through one multivariate method, Student's t-test.

The remainder of this work is organized as follows. Section 2 briefly describes the methodology, including Binary MOGWO, GP, and decision-making techniques. Numerical examples and discussions are provided in Section 3. Finally, research conclusions are listed in Section 4.

2. Methodology

In previous SOH studies, a single evaluation criterion (e.g., RMSE) is usually adopted (Yang et al., 2021; Bercibar et al., 2016). In this work, when applying the wrapper structure for SOH prediction, additional objectives besides regression accuracy are considered, for example, the number of battery performance features.

From the viewpoint of feature selection, battery feature selection is a kind of multiobjective type (i.e., multiobjective feature selection, MOFS (Xue et al., 2016)), where RMSE in regression and the number of features are two main objectives. Moreover, when using GP to generate corresponding regression equations about capacity, its expressional complexity (Gandomi et al., 2015b) is added as a third objective in this study.

In this section, the multiobjective searching method (Binary MOGWO) is described, followed by the evaluating method (GP) and decision-making approaches.

2.1. Searching method-Binary MOGWO

The mechanism of MOGWO is presented first, followed by a brief overview of the modifications to MOGWO to handle binary decision variables.

GWO is a metaheuristic algorithm developed by Mirjalili et al. (2014), inspired by the hunting behaviors of grey wolves in a pack. Grey wolves exhibit different roles based on their hierarchical structure

during hunting activities. There are four categories within the pack, namely, alpha (α), beta (β), delta (δ), and omega (ω). The alpha wolf is the leader, representing the best solution in optimization. The hunting behavior of the wolf pack is guided by α , β , and δ , while the ω wolves follow these three wolves.

To address multiobjective optimization problems, an external archive is incorporated into the original GWO to form MOGWO (Mirjalili et al., 2016). Additionally, two important operations are introduced: the grid mechanism, which aims to enhance the non-dominated solutions in the archive, and the leader selection mechanism, which serves to update the solutions, respectively.

2.1.1. Basics of GWO

Three main phases of grey wolf hunting are mimicked in the GWO algorithms: encircling the prey, hunting the prey, and attacking them.

During the encircling stage, the mathematical model is expressed as follows (Mirjalili et al., 2014):

$$\vec{D} = |\vec{C} \bullet \vec{X}_p(t) - \vec{X}(t)| \quad (1)$$

$$\vec{X}(t+1) = \vec{X}_p(t) - \vec{A} \bullet \vec{D} \quad (2)$$

where t is the current iteration, $\vec{X}(t)$ and $\vec{X}_p(t)$ are the positions of a grey wolf and the prey, respectively. $\vec{A}$ and $\vec{C}$ are vectors, calculated by following equations (Mirjalili et al., 2014):

$$\vec{A} = 2\vec{a} \bullet \vec{r}_1 - \vec{a} \quad (3)$$

$$\vec{C} = 2 \bullet \vec{r}_2 \quad (4)$$

where $\vec{r}_1$ and $\vec{r}_2$ are two random vectors in $[0,1]$, respectively. Elements of $\vec{a}$ vector decreases linearly from 2 to 0 over the whole iteration process.

During the hunting stage, according to the hierarchical structure of the grey wolf pack, it is assumed that the first three best solutions (i.e., α , β , and δ) have better knowledge about the potential locations of the prey. These optimal positions are retained and used to guide other wolves in updating their positions. Mathematically, the following model is adopted (Mirjalili et al., 2014):

$$\vec{D}_\alpha = |\vec{C}_1 \bullet \vec{X}_\alpha - \vec{X}|, \vec{D}_\beta = |\vec{C}_2 \bullet \vec{X}_\beta - \vec{X}|, \vec{D}_\delta = |\vec{C}_3 \bullet \vec{X}_\delta - \vec{X}| \quad (5)$$

$$\vec{X}_1 = \vec{X}_\alpha - \vec{A}_1 \bullet \vec{D}_\alpha, \vec{X}_2 = \vec{X}_\beta - \vec{A}_2 \bullet \vec{D}_\beta, \vec{X}_3 = \vec{X}_\delta - \vec{A}_3 \bullet \vec{D}_\delta \quad (6)$$

$$\vec{X}(t+1) = (\vec{X}_1 + \vec{X}_2 + \vec{X}_3) / 3 \quad (7)$$

The positions of the first three best wolves predict the location of the prey, and other wolves in the pack update their positions randomly around the prey.

The final operation of the grey wolves' hunting behavior is attacking the prey. This procedure called the exploitation phase of GWO, is accomplished by decreasing the $\vec{a}$ vector. When the values of $\vec{A}$ fall into the range of $[-1, 1]$, the next updated position of a wolf will be located between its current position and the prey. This signifies the beginning of the attack operation.

Conversely, searching for potential locations of the prey is another important task for GWO, known as the exploration phase. This mechanism is achieved through $|\vec{A}| > 1$, allowing GWO to find global solutions. An additional auxiliary factor supporting the exploration capability of GWO is the $\vec{C}$ vector, which also aims to avoid local optimal traps.

The GWO algorithm first generates a random population of grey wolves. During the iteration process, α , β , and δ estimate the location of the prey and guide other wolves. The population updates its positions

based on the distance from the prey. The choice between exploration and exploitation depends on the difference between $|\vec{A}|$ and 1. When the predefined termination condition is satisfied, the GWO algorithm terminates.

2.1.2. Extension to MOGWO

Inspired by the mechanism of MOPSO (Coello et al., 2004), two additional mechanisms, i.e., external archive and leader selection, are adopted to handle multiobjective cases. The external archive is employed to store non-dominated Pareto solutions, while leader selection assists the algorithm in choosing the first three superior wolves (i.e., α , β , and δ) as leaders from this external archive.

The key operation concerning the external archive is its management. The entrance control of a new member depends on its Pareto dominance. If the new solution is dominated by a member of the archive, it will be rejected. If it dominates a member of the archive, the dominated solution should be replaced by the new solution. If the new solution and members of the archive do not dominate each other, the solution is added to the archive. If the archive is full, those solutions located in the most crowded segment of the Pareto front are deleted based on the grid mechanism.

In contrast, to merely finding the first three optimal members (i.e., α , β , and δ) in the single-objective optimization case, leader selection in the external archive is performed using a roulette-wheel approach in the least crowded parts of the search area. The selected non-dominated solutions are designated as α , β , and δ wolves.

The position updating mechanism of search agents and the roles of $|\vec{A}|$ are shown in Fig. 1(a), while the pseudo-code of MOGWO is listed in Fig. 1(b) (Mirjalili et al., 2016).

2.1.3. Binary modification

The aforementioned GWO and MOGWO are designed for continuous optimization problems. A binary version of MOGWO for feature selection, in which Boolean values are expressed as selected (1) or unselected (0), is required. The S and V functions are widely employed to convert a continuous version into a binary one in the metaheuristic optimization field (Too and Abdullah, 2020; Song et al., 2021). In this work, the S function is adopted and expressed as follows:

$$S(x) = \frac{1}{1 - \exp(-x)} \quad (8)$$

The position of each search agent in the d^{th} dimension and t^{th} iteration will be transformed through $S(x_i^d(t)) = 1 / [1 - \exp(-x_i^d(t))]$. Therefore, the feature selection statute (0 or 1) is obtained by the following equation:

$$x_i^d(t) = \begin{cases} 0 & \text{if } \text{rand} \leq S(x_i^d(t)) \\ 1 & \text{if } \text{rand} > S(x_i^d(t)) \end{cases} \quad (9)$$

where rand is a uniform random number in $[0,1]$. After each updating operation in Fig. 1(b), Eqs. (8) and (9) are utilized to refine the updated positions. These values are exploited to reach the binary grey wolf position. A binary version of the original GWO (i.e., Binary GWO) and Binary MOGWO are obtained then.

2.2. Evaluating method-GP

GP was introduced by Koza in 1992 (Koza, 1992). It is a symbolic optimization approach. An initial population is generated first, and a new generation is created through genetic operators, such as crossover and mutation. The crossover operator selects a node from the parental individuals and replaces the subtree under the selected nodes, while the mutation operator generates a new node and exchanges another node of a tree. These operations continue until the termination criteria are

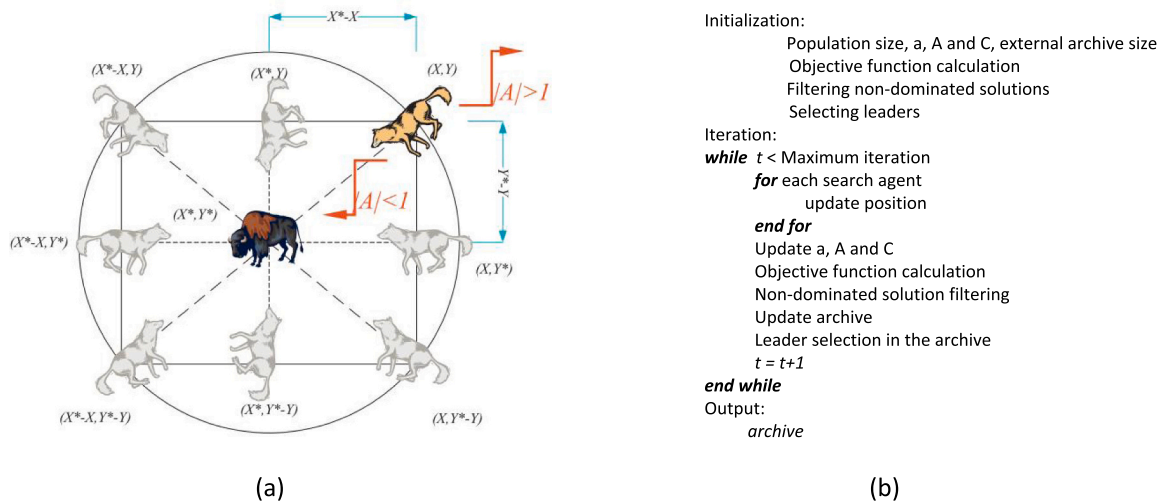


Fig. 1. Position updating of grey wolves, roles of $|A|$ (a) and pseudo code of MOGWO (b) (From (Mirjalili et al., 2016)).

satisfied (Gandomi et al., 2015a).

An example of a symbolic model in GP is shown in Fig. 2(a) (Searson et al., 2010), and the GP flowchart for SOH prediction is given in Fig. 2 (b).

Over the past three decades, many GP codes have been developed and are widely used in various applications (Žegklitz and Pošik, 2020). In this work, the open-source software GPTIPS 2.0 (Searson et al., 2010) is chosen to study SOH prediction. GPTIPS provides several convenient functions for investigating the models (Gandomi and Atefi, 2020), which makes constructing prediction models easier.

GPTIPS receives the feature selection command from Binary MOGWO or Binary GWO and generates its initial population. After predefined termination conditions are met, the optimal model (i.e., expression about SOH prediction and other necessary information) is sent back to Binary MOGWO or Binary GWO for further operations.

2.3. Decision-making method

Typically, the number of non-dominated solutions in the Pareto set is more than one. A necessary decision-making approach is needed to select the final compromise solution within that set. Numerous methods have been developed in engineering applications, such as the utility function (Kim and Choi, 2001) and TOPSIS (Behzadian et al., 2012). In this work, two kinds of decision methods, 'nearest to the utopian

solution' (objective) and preference-based (subjective), are adopted (Parsons, 2009; Ishizaka and Nemery, 2013).

The utopian solution is composed of the corresponding minimal single objective function values. Normalized multidimensional Euclidean distance is employed as a criterion to filter the final solution, as shown in Eq. (10) (Parsons, 2009; Ishizaka and Nemery, 2013):

$$\min |d(x)| = \left\{ \sum_{k=1}^K [(f_k(x) - f_k^o) / f_k^o]^2 \right\}^{0.5} \quad (10)$$

where $|d(x)|$ is the distance from the utopian solution in the Pareto set. k is the k^{th} objective function (K - number of function objectives), f_k^o is the k^{th} optimal single objective function value, while $f_k(x)$ is the k^{th} objective function in the Pareto set obtained by Binary MOGWO. In the present study, three single-objective optimization problems are solved through Binary GWO to find f_k^o .

The decision procedure of 'nearest to the utopian solution' does not consider the subjective information of the designer. On the other hand, the preference-based method provides more freedom for the designer to select the final solution.

The proposed framework in this work is shown in Fig. 3.

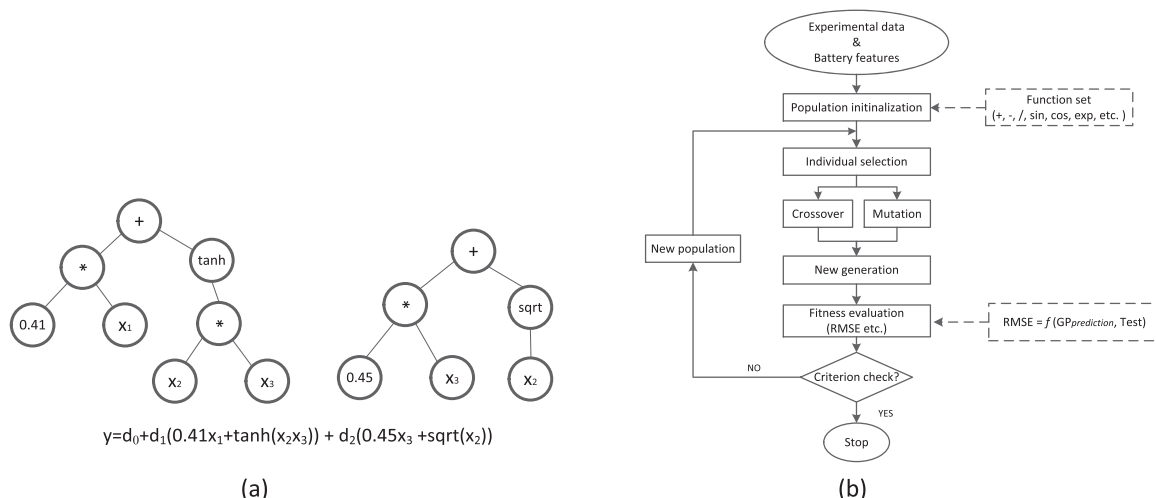


Fig. 2. An example of a symbolic model (a)(From (Searson et al., 2010)) and flowchart of GP for SOH prediction (b).

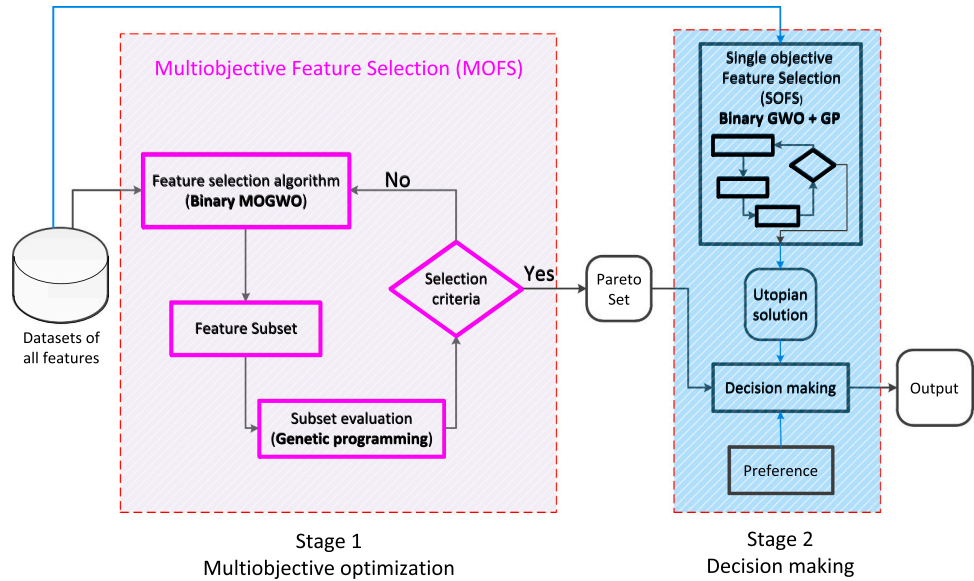


Fig. 3. Flowchart for SOH prediction using MOGWO/GWO and decision making.

3. Numerical examples and discussions

3.1. Experimental data and feature set

As SOH does not correspond to a specific physical quality, there is no industry consensus on how it should be determined. However, compared to internal resistance, impedance, and conductance, the capacity of a battery directly measures its energy storage capability, reflects aging and degradation, aligns with performance standards, impacts safety and reliability, and is easily measured by any battery management system (Shu et al., 2021). Therefore, in this paper, battery capacity is selected as the measure of SOH. To illustrate the proposed flowchart, seven batteries from the NASA Ames Prognostics Center of Excellence (PCoE) (Saha and Goebel, 2007) are chosen. Their cycle conditions are shown in Table 1. Fifteen performance features across four categories for predicting the capacity of these batteries are selected from previous work (Hu et al., 2021a) and are provided in Table 2 (refer to Nomenclature).

3.2. Optimization models

As shown in Fig. 3, two optimization models are developed for SOH prediction in this work.

3.2.1. Multiobjective optimization model

To predict the SOH of a battery with higher accuracy, reduced cost, and a high level of understandability for GP models, three objectives are chosen: prediction accuracy (*RMSE*), the number of features selected (*n*), and the expressional complexity of the GP result.

Suppose *x* is a 15-dimensional decision variable, reflecting the selection status (0 or 1) of each feature in Table 2. The objective functions are shown in Eq. (11):

Table 2
Performance features for prediction of SOH (Hu et al., 2021a).

	Category	Features
1	Voltage-based	VIECT, ICHV, VDEDT, IDV
2	Time-based	TEVI, TEVD, TECD, TCCC, TCVC
3	Temperature-based	MATC, MATCL, MATD, MATDL, MET
4	Current-based	CDECT

$$\min f(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), f_3(\mathbf{x})) \quad (11)$$

where $f_1(\mathbf{x}) = RMSE$, $f_2(\mathbf{x}) = n$ and $f_3(\mathbf{x}) = Complexity$. *RMSE* is obtained from (Fei et al., 2021)

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (12)$$

where y_i is the observation value, $\hat{y}_i$ is the prediction values through GP, *N* is the number of testing samples. *n* and *Complexity* are the total feature used and expressional complexity of the GP models, respectively.

3.2.2. Single objective optimization model

To find the utopian solution, three single objective optimization models are employed. That is, $f_1(\mathbf{x})$, $f_2(\mathbf{x})$, and $f_3(\mathbf{x})$ are selected as objective independently to find their corresponding optimal values, f_1^o , f_2^o and f_3^o (Eq. (10)), respectively.

According to the mechanism of GP, the final capacity estimation formula will not be unique at the end of the search process. To ensure the robustness of the GP result, 20 independent runs are employed to find the final optimal results.

Table 1
Cycle conditions of four batteries from NASA PCoE.

Battery label	Charging cut-off voltage (V)	Discharging cut-off voltage (V)	Charging constant current (A)	Discharging current (A)	Temperature (°C)
B05	4.2	2.7	1.5	2	24
B06	4.2	2.5	1.5	2	24
B07	4.2	2.2	1.5	2	24
B18	4.2	2.5	1.5	2	24
B33	4.2	2.0	1.5	4	24
B34	4.2	2.2	1.5	4	24
B36	4.2	2.7	1.5	2	24

Table 3
Parameters in optimization codes and GPTIPS 2.0.

Codes	Parameters	Values
Binary MOGWO	Population size	100
	Max iteration	100
	Archive size	100
	Grid inflation parameter	0.1
	Number of grids per each dimension	30
Binary GWO	Leader selection pressure parameter	4
	Population size	100
	Max iteration	300
GPTIPS 2.0	Population size	30
	Max iteration	50
	Tournament size (in selection)	6
	Termination (fitness value), less than	0.0001
	Training set vs. testing set	70:30
	Genes size	6
	Function nodes	×, +, −, /, sqrt, square, sin, cos, exp

The individual with the best *RMSE* value in the final population of GP and corresponding expressional complexity will be defined as the evaluation result and sent back to Binary MOGWO. *RMSE* of each solution is calculated through the regression equations of GP and the experimental data of the battery.

The objective function space of the Pareto set for the B05 battery is shown in Fig. 4. 50 discrete solutions are found by Binary MOGWO. It is observed that Pareto solutions are relatively well-distributed in the objective function space, providing more choices for designers. This is due to the powerful searching capability of Binary MOGWO.

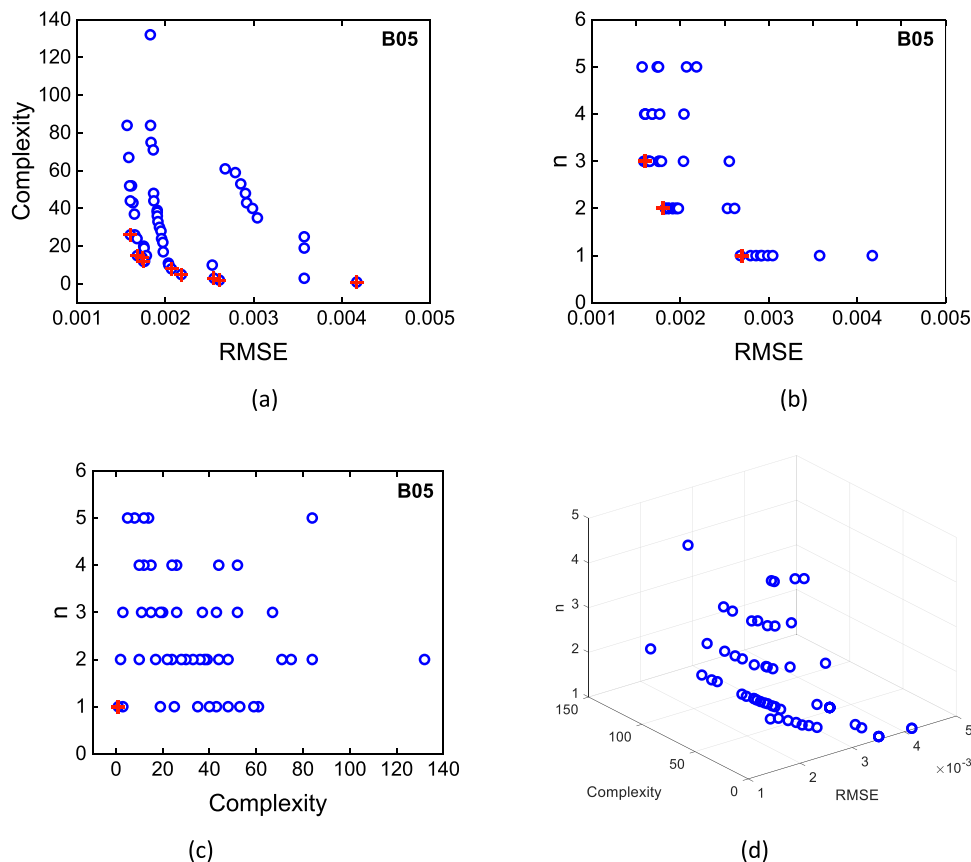


Fig. 4. Pareto set plots of B05 battery.

Conflicting characteristics among solutions are found. One cannot find a solution with optimal values in all three objective functions concurrently. This is the inherent performance of the Pareto solution. Therefore, a necessary trade-off between objective functions should be considered during the filtering of the final compromise solution.

Designers often choose the solution through the selection of acceptable objective performance, as shown by the Pareto front (Coello et al., 2007). In 2D plots, those solutions that are located at the Pareto front are shown as red crosses (+). Battery designers may have much interest in selecting red-cross solutions in Fig. 4. For example, one may choose $n=2$ or $n=3$ solutions in Fig. 4(b) for a lower number of features with higher *RMSE*.

In the Pareto set those solutions with only one feature and different complexity are worth noting. For example, 10 Pareto solutions with $n=1$ (*MATDL*) are filtered in Fig. 4. Their *RMSE* –complexity and predictions errors are shown in Fig. 5, while two solutions with lowest and highest complexity are given in Table 4.

It is suggested that the proposed framework can (1) select suitable feature(s) in the feature sets and (2) give a concise expression of the SOH with the help of GP. The intrinsic relationship between features (*MATDL*) and the *Capacity* is also mined then.

3.3. Decision making

In practical applications, solving multi-objective optimization problems is usually accompanied by a decision-making process. According to the definition of multi-objective optimization solutions, each solution in the Pareto set can be chosen as the final compromise one. In this section, the 'nearest to the utopian solution' method is utilized to find the compromise solution for the B06 battery first (shown in Fig. 6), followed by another preference-based decision-making method.

In Fig. 6, Points A, B, and C are single-objective optimization results

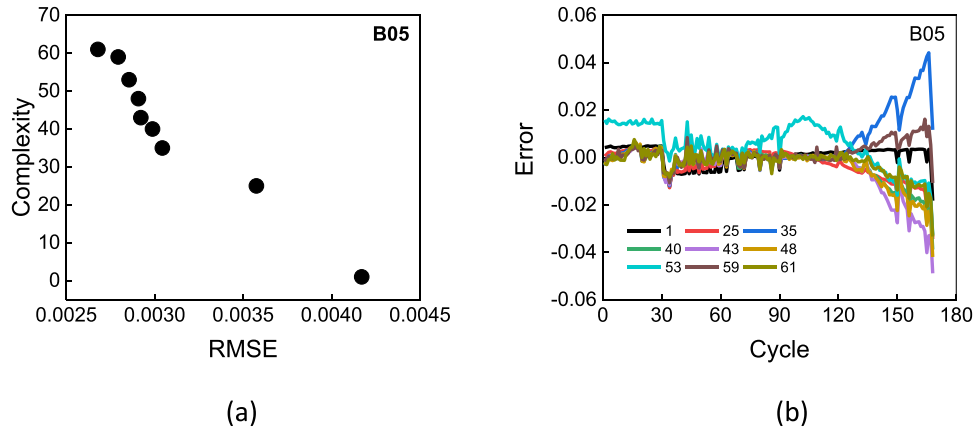


Fig. 5. Pareto solutions of B05 with $n=1$ (MATDL) (a) and prediction error for different complexities (b).

Table 4

Two Pareto solutions only based on MATDL with different complexities (B05).

RMSE	Complexity	Expression
0.00268	61	$Capacity = 0.006228 \times \cos(MATDL^{0.5} + 0.9725) - 0.004554 \times \cos(0.9049 \times MATDL^{0.5}) - 0.004123 \times \sin(0.6704 \times MATDL) - 0.001021 \times \sin(0.8188 \times MATDL) + 0.06002 \times MATDL^{0.5} - 1.623$
0.00417	1	$Capacity = 0.0005529 \times MATDL - 0.0008818$

for $\min f_1(\mathbf{x})$ ($= 0.0035515$), $\min f_2(\mathbf{x})$ ($= 2$) and $\min f_3(\mathbf{x})$ ($= 1$), respectively. Therefore, the objective function of the utopian solution is (0.0035515, 2, 1).

The results of single-objective optimization are provided in Table 5. It is found that, to find the minimal MSE, 9 features are employed and more items are needed in the GP expression, leading to higher expressional complexity ($= 133$). The use of more features in the prediction expressions requires more labor and incurs higher analysis costs. If fewer features are used, RMSE will increase then, about 23 % for $\min f_2(\mathbf{x})$ and 301 % for $\min f_3(\mathbf{x})$, respectively.

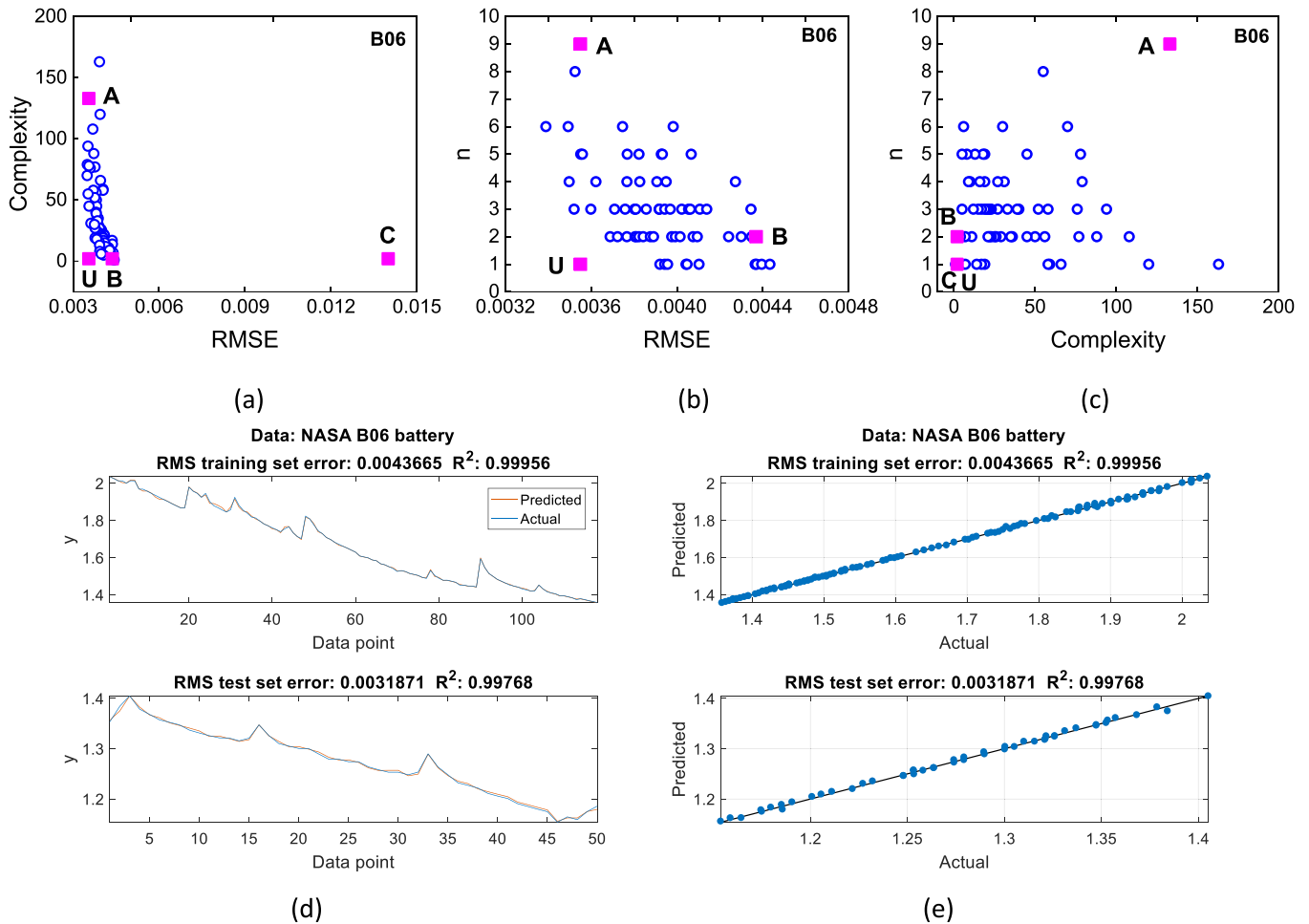


Fig. 6. (a)-(c) Pareto set of B06; (d)- training and testing error in GP; (e)- B06 SOH prediction and error.

Table 5
Single optimization results of B06 battery.

Objective function	RMSE, <i>n</i> , complexity	Features selected	Capacity expression	Point in Fig. 5
$\min f_1(x)$	0.0035515 9 133	MATDL, TCVC, MATC, MET, MET, CDECT, TCCC, ICHV, TEVD	$\text{Capacity} = 0.000545 \times \text{MATDL} - 1.028 \times 10^{-9} (\text{TCVC} + \text{MATC} + \text{MET})(\text{TCVC} + \text{MATC} + \text{MET} + \text{CDECT}) - 7.383 \times 10^{-7} \times \text{TCCC} \times \cos(\text{ICHV}^2 \times \text{MATC}) - 2.217 \times 10^{-10} \times \cos(\text{TCVC} + \text{MATC} + \text{MET}) \times (\text{TEVD} + \text{MATCL} + \text{MET} - \text{CDECT}) - 4.232 \times 10^{-11} \times \text{TCVC} \times \cos(\text{TCVC}) \times (\text{TEVD} + \text{TCVC} + \text{MET}) + 1.0373 \times 10^{-10} \times \text{TCVC} \times \text{MATDL} \times \text{MET} + 2.976 \times 10^{-7}$	A
$\min f_2(x)$	0.0043665 2 2	IDV MATDL	$\text{Capacity} = 0.137 \times \text{IDV} + 0.0005595 \times \text{MATDL} - 0.6008$	B
$\min f_3(x)$	0.0142420 1 1	TEVD	$\text{Capacity} = 0.0004536 \times \text{TEVD} + 1.018$	C

The final compromise solution is obtained according to the normalized multidimensional Euclidean distance, and in this example, it is just the B point. Solution B has smaller expressional complexity and only two features, with a relatively larger RMSE value (Fig. 6(b)).

In the Pareto set (Fig. 6(a-c)), the bounds of RMSE and *n* are much smaller than that of Complexity. The performance of Complexity will dominate the location of the B point, i.e., approaching lower the Complexity while RMSE value is not the main concern in decision-making. Therefore, a larger RMSE value is also acceptable. This decision result can be explained through the entropy weight (Behzadian et al., 2012) of these three performances in the Pareto set, as shown in Table 6. It is demonstrated that the roles of RMSE for B05 and B06 in the decision process are much smaller.

The ‘nearest to the utopian solution’ method for B06 yields a concise expression ($n=2$, complexity=2) for SOH with a relatively higher RMSE. However, this solution is not at the Pareto front of RMSE – *n* plot (refer to Fig. 6(b)). This is an intrinsic characteristic of a Pareto solution, i.e., not all objective function values of a solution will be optimal in the set.

On the other hand, preference-based decision-making is also frequently utilized in practical design. If one prefers to select a compromise solution just on the Pareto front, a new attempt to find another final compromise solution can be achieved by slightly increasing the feature number. For example, $n = 3$ is selected for the B05 battery in Fig. 4(b), and the expression of GP is listed in Eq. (15):

$$\begin{aligned} \text{Capacity} = & 0.0009122 \times \text{MATDL} - 0.06409 \\ & \times \text{TEVD} + 1.789 \cos^2(\text{IDV}) - 0.0005087 \times \text{TEVD} \\ & \times \cos^2(\text{IDV}) - 0.00134 \times \text{MATDL} \times \cos^2(\text{IDV}) + 0.01533 \\ & \times \text{IDV} \times \text{TEVD} - 0.4764 \end{aligned} \quad (15)$$

The feature TEVD is added and expressional complexity increases to 37, while RMSE= 0.0016490. Compared with $n= 2$, this solution only increases RMSE slightly and is acceptable.

However, although the proposed method offers competitive results for SOH estimation by finding the optimal feature subset, there are still some considerations worth noting. MOGWO can exhibit slow convergence, particularly in complex optimization landscapes with multiple local optima. The convergence speed is influenced by the balance between exploration and exploitation phases, which can be challenging to optimize, thus leading to no guarantee of finding the global solution. Additionally, in GP, initialization bias can significantly influence the evolutionary process, often resulting in excessively large and complex expressions without a corresponding improvement in fitness. Therefore,

Table 6
Entropy weight of three objective functions for B05 and B06.

	RMSE	Complexity	<i>n</i>
B05	0.0691	0.7294	0.2016
B06	0.0033	0.7840	0.2128

multiple simulations are needed when combining GWO and GP to enhance exploration and exploitation, ensure robustness and reliability, mitigate the risk of local optima, and evaluate diverse feature sets, thus overcoming the drawbacks of each method and achieving more accurate and reliable solutions.

3.4. Further discussions

3.4.1. An alternative model

Solution B in Fig. 5 (B06) and Eq. (15) (B05) are obtained from their cycling data and corresponding decision-making procedures, respectively. One of the main concerns in the present work is to construct a concise expression for the SOH of a battery set. In addition to the expression features of Solution B and Eq. (15), another constrained single-objective optimization is studied here. All cycling data of B05, B06, B07, and B18 are considered together. $n = 3$ is required for the sake of lower cost, and only RMSE is set as the optimal objective.

The optimization model is expressed as follows:

$$\begin{aligned} \min f(x) = & \text{RMSE} \\ \text{Subjected to } & n = 3 \end{aligned} \quad (16)$$

Then, a new expression of battery capacity is obtained through constrained single objective optimization solved by Binary GWO with GP, shown in Eq. (17):

$$\begin{aligned} \text{Capacity} = & 0.03852 \times \text{MATDL} - 0.01952 \times \text{TEVD} + 0.002869 \\ & \times \exp(2 \times \text{IDV}) - 0.1892 \times \exp(\text{IDV}) - 0.009064 \times \text{IDV} \\ & \times \text{MATDL} + 0.01113 \times \text{IDV}^2 \times \text{TEVD} - 0.005963 \end{aligned} \quad (17)$$

Regression results are $\text{RMSE} = 1.6904 \times 10^{-3}$ and $\text{Complexity} = 27$, respectively. The constraint handling is based on the death penalty function method (Mirjalili et al., 2014). When a solution violates any constraints, a penalty factor will be employed to form a big added value (for the minimization problem) to the original objective function.

To examine the applicability of Solution B, Eq. (15) and Eq. (17), prediction results and corresponding error curves are shown in Fig. 7. It is observed that the prediction accuracy of these three equations is acceptable.

3.4.2. Comparison with other works

The above three expressions are compared with the results from single-objective optimization (RMSE) in the literature, listed in Table 7. It is found that the prediction accuracy of this work is better than those of the literature for these three batteries. It is also worth noting that Solution B in Table 4 (B06) and Eq. (15) (B05) yield the best prediction accuracy in their datasets, respectively. When these two expressions are applied to other batteries, RMSE will increase slightly.

The Pareto sets of B05 and B06 suggest that just one feature, the MATDL in the NASA battery dataset, can be utilized to construct the

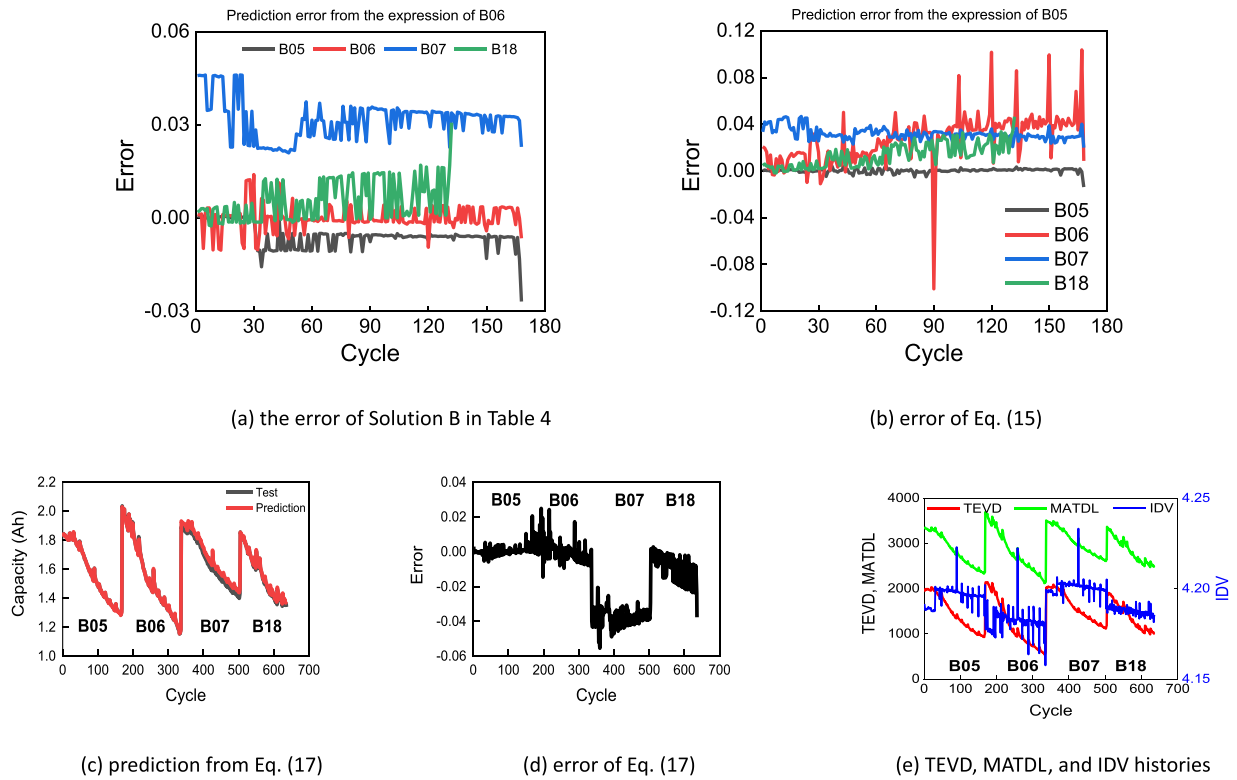


Fig. 7. Application of three equations for four batteries.

Table 7
RMSE comparisons.

Battery label	(Yao et al., 2020)		This work		
	Features	RMSE	Features	RMSE	Equations
B05	g_1, g_3, g_4	3.827E-3	IDV, MATDL	2.680E-3	Solution B in Table 4
			IDV, MATDL, TEVD	1.650E-3	Eq. (15)
			IDV, MATDL, TEVD	4.382E-3	Eq. (17)
B06	g_1, g_3	6.628E-3	IDV, MATDL	4.366E-3	Solution B in Table 4
			IDV, MATDL, TEVD	6.766E-3	Eq. (15)
			IDV, MATDL, TEVD	7.526E-3	Eq. (17)
B07	g_2, g_3, g_4, g_5	4.271E-3	IDV, MATDL	3.225E-2	Solution B in Table 4
			IDV, MATDL, TEVD	4.138E-3	Eq. (15)
			IDV, MATDL, TEVD	3.528E-2	Eq. (17)

Note: g_1 —time interval of charging voltage 2.7 V rising to 4.2 V; g_2 —time interval in CV discharge current 1.5 A to 0.3 A; g_3 —time interval discharge voltage reduced from 3.7 V to 2.7 V; g_4 —average battery temperature; g_5 —average discharge temperature (Yao et al., 2020).

prediction equation. Another three batteries, B33, B34, and B36 with different test conditions are also examined here. The corresponding expressions (via constrained optimization) and RMSE comparisons with (Shui et al., 2018) (with one feature, g_3 used in the expression) are provided in Table 8.

The RMSE of B33 and B34 is slightly better than those of (Yao et al., 2020). However, the expression is more complex. Due to the two abnormal test data in B36 (Yao et al., 2020), a larger RMSE of B36 is found then.

Table 8
Results of B33, B34 and B36.

Battery label	(Yao et al., 2020)	This work		Capacity expression ($x = MATDL$)
	RMSE	RMSE		
B33	0.0017	0.0016953	$0.001066x + 0.01731\cos(0.0206x) + 0.009480\sin(\sin(9.92 \times 10^{-3}x^2))/(9.92 \times 10^{-3}x + \cos(-0.003194x\cos(-0.003194x))) - 0.06216$	
B34	0.0019	0.0018380	$0.0012276x + 0.01484\cos(6.2265 + 5.223x^2 - 14320.135x) + 0.0001233x\cos(3.996 + 5.2231x^2 - 14320.26x) + 0.007617\cos(1.42471 + 486.42x + 0.14397\cos(6.2265 + 5.2231x^2 - 14320.135x)) - 0.29334 - 0.17759\cos(3.996 + 5.2231x^2 - 14320.26x)$	
B36	0.0297	0.056613	$1.9056 + 0.0009945x\sin^2(3.809 \times 10^{-7}x^2) + 1.7319e - 14x^4\sin^2(3.809 \times 10^{-7}x^2) - 0.000112x - 0.2984\sin(3.809 \times 10^{-7}x^2) - 4.8490 \times 10^{-7}x^2\sin^2(3.809 \times 10^{-7}x^2)$	

It is worth noting that g_3 and MATDL are both time intervals related to the discharging process. However, g_3 is from the voltage curve, and MATDL is obtained through the temperature source. It implies that the voltage and temperature connection in the lithium discharging procedure can be captured by GP. It is also suggested that, compared with specifying battery features for SOH directly, the wrapper structure has the potential for finding more prediction expressions using the feature set with comparable accuracy. It is also anticipated that different features may lead to different expressional complexity, depending on the intrinsic relationship between the feature and capacity performance.

Solution B (B06) is a linear combination IDV and MATDL. It has a simpler structure than those Eqs. (15) and (17) obtained in this work.

This prediction structure is extended to the applications of B33, B34, and B36. In Eq. (18) c_1 , c_2 and c_3 are determined through GWO.

$$\text{Capacity} = c_1 \times IDV + c_2 \times MATDL + c_3 \quad (18)$$

Three coefficients and corresponding RMSE are given in Table 9. Eq. (18) can provide acceptable accuracy (<2.5 %) for B33 and B34 with 2 features. A slightly larger RMSE of B36 is also dependent on those two abnormal monitoring data (Yao et al., 2020). The validation of linear combination style in IDV and MATDL is confirmed.

Also, to comprehensively evaluate the proposed methodology, a comparative analysis with both conventional and state-of-the-art SOH prediction methods is conducted in Table 10.

The results indicate that our proposed method achieved superior accuracy and reliability, which may be attributed to the combination of GP with MOGWO. This integration theoretically offers a more balanced solution through multiple runs with different subsets of input features.

3.4.3. Data mining

According to the proposed framework, only two or three performance features (Table 7 & 9) remained when GP is finished. This is due to three main factors:

- (1) Higher Pearson Correlation (Härdle and Simar, 2019) between MATDL, TEVD and the Capacity (shown in Table 11);
- (2) Larger effect size of MATDL and TEVD upon the Capacity;
- (3) The higher frequency of these features in the population and the intrinsic mechanism of GP. Higher frequency means more chances to be chosen by the selection and mutation operations.

It is worth to be noted that a lower correlations feature, IDV is also included in the final population. This is due to the optimization selection pressure and higher searching ability of the GP method. However, the effect of IDV upon the battery capacity is small. The main variation trendy of the Capacity in Solution B is controlled by MATDL, as shown in Fig. 8 (constant item −0.6008 is omitted) and TEVD (Fig. 7(e)).

To investigate the effects of all 15 features upon the Capacity, a statistical measurement, Student t-test is employed here for B05, B06, B07, and B18 (Esteco, 2017). The effect size is obtained through the following procedures:

Step 1 Dividing the input vector (15 features) into the lower part and upper part, according to its mean value.

Step 2 Calculating the mean value of Capacity (M_- and M_+) in the corresponding lower and upper parts of this input variable, respectively.

Step 3 $EffectSize = M_+ - M_-$

A positive effect value reflects a direct relationship while a negative value shows an inverse relationship. Effects sizes of B05, B06, B07, and B18 are shown in Fig. 9.

It is observed that IDV features have smaller effect sizes while MATDL and TEVD have higher positive effect sizes. This result is consistent with the variation tendency in Fig. 8.

3.5. Summary

This paper presents a two-stage prediction framework for accurately predicting the State of Health (SOH) of lithium-ion batteries using metaheuristic algorithms and genetic programming (GP). The framework aims to balance three objectives: minimizing the root-mean-square error (RMSE) of predictions, reducing the number of selected features, and keeping the expression complexity low. The framework is validated using the NASA Prognostics Center of Excellence battery dataset,

Table 9
Applications of linear prediction structure for B33, B34, and B36.

Battery label	c_1	c_2	c_3	RMSE
B33	0.13945	0.0010696	-0.65458	0.025232
B34	0.16837	0.0011681	-0.91962	0.020628
B36	1.28350	-3.921192	-3.56021	0.068257

Table 10
RMSE comparison with other method.

Battery label	LSTM (He et al., 2024)	PSO-LSTM (He et al., 2024)	CG-GPR (He et al., 2024)	PSO-GPR (He et al., 2024)	SVM (Zhao et al., 2022)	Present (Table 7)
B05	0.044	0.04	0.0112	0.0069	0.03	0.0017
B06	0.03	0.036	0.022	0.013	0.07	0.0044
B07	0.055	0.019	0.016	0.0078	–	0.0041

Table 11
Pearson Correlation and p -value between extracted features and the battery capacity.

Features	Item	Battery label			
		B05	B06	B07	B18
IDV	Correlation	-0.2186	0.0522	0.0263	0.6163
	p -value	0.004	0.005	0.003	0.000
MATDL	Correlation	0.9995	0.9998	0.9998	0.9995
	p -value	0.000	0.000	0.000	0.000
TEVD	Correlation	0.9984	0.9945	0.9983	0.9982
	p -value	0.000	0.000	0.000	0.000

demonstrating its ability to provide SOH predictions in symbolic equation form with high accuracy, low cost, and reduced complexity. The study contributes to the field by proposing a novel integration of multiobjective optimization and decision-making techniques for SOH prediction, offering deeper insights into the relationships between selected features and battery capacity.

4. Conclusions

In pursuit of developing an explicit and concise prediction expression for lithium-ion battery SOH, a two-stage framework has been proposed. The Pareto set for the multiobjective optimization problem is determined using Binary MOGWO, while the final compromise solution is obtained through essential decision-making approaches. Seven NASA battery examples are employed to demonstrate the effectiveness of the proposed framework.

Based on the study, several conclusions can be drawn:

- (1) The proposed two-stage framework, which combines multi-objective optimization and decision-making approaches, is effective in finding concise expressions for SOH with higher regression accuracy and lower cost.
- (2) The wrapper structure employed in both multi- and single-objective feature selection procedures has the potential to discover additional SOH expressions using different features.
- (3) The Pareto set of SOH predictions identified by Binary MOGWO is well-distributed, offering more options for designers. Binary GWO also demonstrates a strong ability to find solutions in the constrained decision variable space of features.
- (4) Correlation analysis and t-tests provide further insight into SOH expressions. Features with higher capacity-related correlations and larger effect sizes are more likely to be present in the final expression.

Also, several limitations in this study should be considered:

- (1) Only seven NASA batteries were chosen for examination, all of which have the same chemical composition and operating conditions, also a total of 15 health indicators were collected from four traditional measurement sources; however, features from calculated sources (e.g., incremental capacity-based (Hu et al., 2021a)) were not included in the feature set.

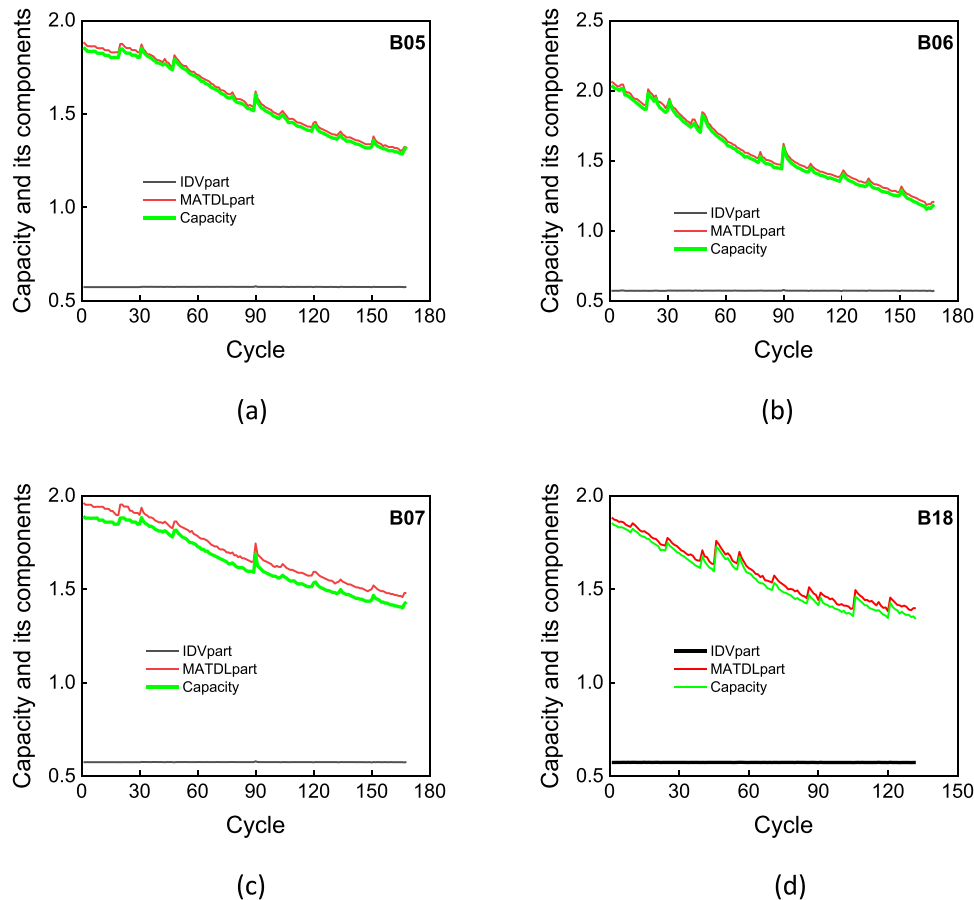


Fig. 8. Component of capacity in Solution B for four batteries.

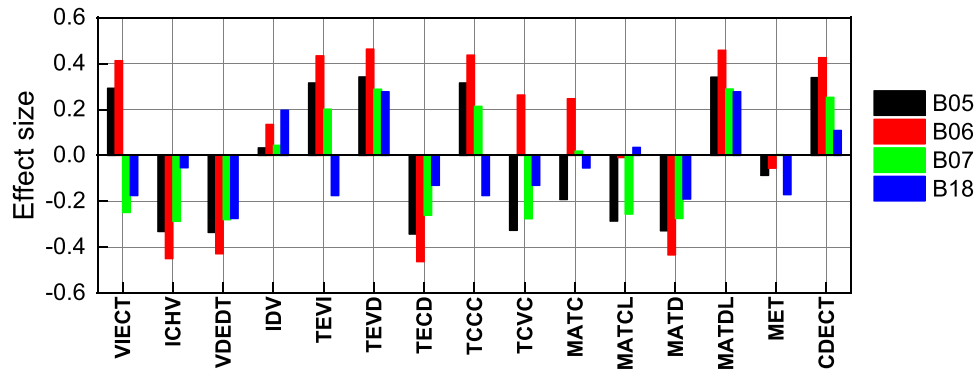


Fig. 9. Effect sizes of all 15 features on the battery capacity.

- (2) GP often lacks flexibility due to its reliance on predefined tree structures and specific genetic operators, which can limit the types of solutions it can evolve. The tree-based representation inherently restricts the variety of program structures, making it challenging to adapt to different problem domains or accommodate complex interactions between variables.
- (3) Also Binary GWO might converge prematurely to a local optimum if the exploration phase is insufficient or if the exploitation phase dominates too early. Additionally, the randomness in the search process can lead to inconsistent results across different runs, further reducing the likelihood of consistently finding the global solution.
- (4) Constraints among the features were not considered. In the multiobjective optimization model, addressing redundancy in the feature set is necessary to reduce labor costs as much as possible.

Future research directions might include:

- (1) Investigating more battery datasets and features using multi-objective optimization and GP.
- (2) Exploring additional metaheuristic optimization algorithms to find the Pareto solution for SOH in the constrained decision variable space.

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Ethical Statement

All data used in this study have been collected and processed in compliance with ethical standards, ensuring data privacy and confidentiality.

Efforts were taken to ensure the datasets used were representative, aiming to minimize biases that could influence the model’s outcomes. This research was conducted with transparency in methodology to allow for reproducibility and scrutiny. Any potential consequences or misuse of the model’s application have been considered and discussed in the [appropriate section/subsection] of this paper.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

A, Nomenclature

Abbreviation	Definition
ANN	Artificial neural network
CC-CV	Constant-current and constant-voltage charge
ECM	equivalent circuit models
EM	electrochemical model
GA	Genetic algorithm
GP	Genetic programming
GPR	Gaussian process regression
GWO	Grey wolf optimization
ICA	Incremental capacity analysis
MAPE	Mean absolute percentage error
MOFS	Multiobjective feature selection
MOGWO	Multiobjective grey wolf optimization
MOPSO	Multiobjective particle swarm optimization
NASA	National Aeronautics and Space Administration
OCP	Open circuit potential
RMSE	Root-mean-square error
RVM	Relevance vector machine
SOFS	Single optimization feature selection
SOH	State of health
SVM	Support vector machine

B, Features set (Hu et al., 2021a)

No	Abbreviation	Definition
1	CDECT	The current decrease during equal charge time
2	ICHV	Initial charge voltage
3	IDV	Initial discharge voltage
4	MATC	The maximum temperature (charge process)
5	MATCL	Location of the maximum temperature (charge process)
6	MATD	The maximum temperature (discharge process)
7	MATDL	Location of the maximum temperature (discharge process)
8	MET	The mean temperature
9	TCCC	The time of constant current charge
10	TCVC	The time of the constant voltage charge
11	TECD	The time interval during the equal current decrease
12	TEVD	The time interval of equal voltage decrease
13	TEVI	The time interval during equal voltage increase
14	VDEDT	Voltage decreases during equal discharge time
15	VIECT	Voltage increases during equal charge time

Data Availability

Data will be made available on request.

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CRediT authorship contribution statement

Li Xuebin: Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation. **Jin Zhao:** Writing – review & editing, Writing – original draft, Supervision, Data curation, Conceptualization. **Li Shengqun:** Software, Methodology, Conceptualization. **Yu Daiwei:** Writing – original draft, Data curation. **Zhang Jun:** Visualization, Data curation. **Zhang Wenjin:** Writing – review & editing, Validation.

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